

From Lie groups to the universal enveloping algebra

Associativity, infinitesimal generators, and noncommutative polynomials

Omar Corona Tejeda
omarct1989@ciencias.unam.mx
The Mathematical Physics Project

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Abstract

The word *algebra* does not imply associativity: the Lie bracket satisfies Jacobi, but in general is not associative, even though the group product is. Using $SO(3)$ as an example, we distinguish these operations and explain how infinitesimal rotations lead to finite rotations through the exponential. We construct $U(\mathfrak{g})$ from the tensor algebra: its commutator, not its product, reproduces the original bracket. We study the universal property, state Poincaré-Birkhoff-Witt, and distinguish free polynomials $\mathbb{K}\langle x_1, \dots, x_n \rangle$, commutative polynomials $\mathbb{K}[x_1, \dots, x_n]$, and rational functions $\mathbb{K}(x_1, \dots, x_n)$. The examples and the Leibniz rule for the commutator complete the discussion and open a connection with quantum mechanics.

1 The initial subtlety: what “algebra” means

In many elementary courses, the word *algebra* is used as shorthand for “unital associative algebra.” This convention is useful, but it is not the broadest definition. In general, an algebra over a field $\mathbb{K}$ is a vector space equipped with a bilinear operation.

Definition 1 (Not necessarily associative algebra). An algebra over $\mathbb{K}$ is a vector space A together with a bilinear map

$$A \times A \longrightarrow A, \quad (x, y) \longmapsto xy. \quad (1)$$

The additional identities satisfied by this product determine the type of algebra: associative, commutative, Lie, Jordan, and so on. With additional structure, one also obtains, for example, Banach algebras and C^* -algebras.

Thus, “being an algebra” does not by itself require that

$$(xy)z = x(yz). \quad (2)$$

Associativity is an additional condition.

Key idea

A useful rule of thumb is

$$\boxed{\text{algebra} = \text{vector space} + \text{bilinear product.}} \quad (3)$$

One must then ask which identities that product satisfies.

2 Lie algebras

Definition 2 (Lie algebra). A Lie algebra over $\mathbb{K}$ is a vector space $\mathfrak{g}$ equipped with a bilinear map

$$[-, -] : \mathfrak{g} \times \mathfrak{g} \longrightarrow \mathfrak{g}, \quad (4)$$

satisfying:

1. alternation: $[x, x] = 0$ for every $x \in \mathfrak{g}$;
2. the Jacobi identity:

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0. \quad (5)$$

Proposition 1 (Alternation and antisymmetry). *Let $[-, -] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$ be a bilinear map. If $\text{char}(\mathbb{K}) \neq 2$ ¹, then*

$$[x, x] = 0 \quad \text{for every } x \in \mathfrak{g} \quad \Longleftrightarrow \quad [x, y] = -[y, x] \quad \text{for all } x, y \in \mathfrak{g}. \quad (6)$$

Proof of Proposition 1. First assume alternation. By bilinearity,

$$0 = [x + y, x + y] = [x, x] + [x, y] + [y, x] + [y, y] = [x, y] + [y, x], \quad (7)$$

so $[x, y] = -[y, x]$. This direction requires no assumption on the characteristic. Conversely, if the bracket is antisymmetric, then $[x, x] = -[x, x]$ and hence $2[x, x] = 0$. When $\text{char}(\mathbb{K}) \neq 2$, the scalar 2 is invertible, so $[x, x] = 0$. This yields alternation. □

The bracket is the product of the Lie algebra. In general, there is no second associative product hidden inside $\mathfrak{g}$. Expressions such as xy , x^2 , or xyz are not defined in $\mathfrak{g}$ unless the algebra comes with additional structure.

The Jacobi identity can also be written as

$$[x, [y, z]] = [[x, y], z] + [y, [x, z]]. \quad (8)$$

This says that, for each $x \in \mathfrak{g}$, the operator

$$\text{ad}_x : \mathfrak{g} \longrightarrow \mathfrak{g}, \quad \text{ad}_x(y) := [x, y], \quad (9)$$

acts as a derivation of the bracket.

2.1 A two-dimensional nonabelian example

Let $\mathfrak{g}$ be the vector space over $\mathbb{K}$ spanned by x and y :

$$\mathfrak{g} := \text{span}_{\mathbb{K}}\{x, y\} \quad (10)$$

with bracket determined by

$$[x, y] := y, \quad [y, x] = -y, \quad [x, x] = [y, y] = 0. \quad (11)$$

The elements of $\mathfrak{g}$ are only linear combinations $ax + by$. There are no degree-two elements such as x^2 or xy .

¹The characteristic of $\mathbb{K}$ is the smallest $n > 0$ with $n \cdot 1_{\mathbb{K}} = 0$, or 0 if no such n exists.

For

$$u = ax + by, \quad v = cx + dy, \quad (12)$$

bilinearity gives

$$\begin{aligned} [u, v] &= [ax + by, cx + dy] \\ &= ac[x, x] + ad[x, y] + bc[y, x] + bd[y, y] \\ &= (ad - bc)y. \end{aligned} \quad (13)$$

The result again lies in the same two-dimensional space.

This bracket is not associative. Indeed,

$$[[x, x], y] = 0, \quad [x, [x, y]] = [x, y] = y. \quad (14)$$

Therefore,

$$[[x, x], y] \neq [x, [x, y]]. \quad (15)$$

Warning

Saying that a Lie algebra is “nonassociative” means that associativity is not one of its axioms. It does not mean that it is forbidden to hold accidentally. In an abelian Lie algebra, where $[x, y] = 0$ for all x, y , both sides of any associative identity are zero.

3 Central example: the rotation group $SO(3)$

The group

$$SO(3) := \{R \in M_3(\mathbb{R}) : R^T R = I, \det R = 1\} \quad (16)$$

describes the oriented rotations of $\mathbb{R}^3$. Its product is matrix multiplication and is therefore associative.

Its Lie algebra is

$$\mathfrak{so}(3) := \{A \in M_3(\mathbb{R}) : A^T = -A\}, \quad (17)$$

with bracket given by the matrix commutator

$$[A, B] := AB - BA. \quad (18)$$

3.1 Explicit infinitesimal generators

The rotations about the three coordinate axes are

$$R_x(\theta) := \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \quad R_y(\theta) := \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}, \quad (19)$$

$$R_z(\theta) := \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (20)$$

The infinitesimal generators are obtained by differentiating at $\theta = 0$:

$$J_i := \left. \frac{dR_i(\theta)}{d\theta} \right|_{\theta=0}. \quad (21)$$

Explicitly,

$$J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \quad (22)$$

Finite rotations are recovered through the matrix exponential:

$$R_i(\theta) = e^{\theta J_i}. \quad (23)$$

For a constant angular velocity $\boldsymbol{\omega} = (\omega_x, \omega_y, \omega_z)$, the corresponding generator is

$$\hat{\boldsymbol{\omega}} := \omega_x J_x + \omega_y J_y + \omega_z J_z = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}. \quad (24)$$

The map $\boldsymbol{\omega} \mapsto \hat{\boldsymbol{\omega}}$ is the standard identification between $\mathbb{R}^3$ and $\mathfrak{so}(3)$. To connect this identity with the physical interpretation, write the position vector as $\mathbf{r} = (r_x, r_y, r_z)^\top$. Matrix multiplication gives

$$\hat{\boldsymbol{\omega}} \mathbf{r} = \begin{pmatrix} \omega_y r_z - \omega_z r_y \\ \omega_z r_x - \omega_x r_z \\ \omega_x r_y - \omega_y r_x \end{pmatrix} = \boldsymbol{\omega} \times \mathbf{r}, \quad (25)$$

which is exactly the component formula for the cross product. Thus, the matrix $\hat{\boldsymbol{\omega}}$ represents the linear operation “taking the cross product with $\boldsymbol{\omega}$.”

3.2 From small rotations to a finite rotation

Intuitively, a finite rotation can be constructed by accumulating many rotations through very small angles about the same axis. Consider a pure rotation about a fixed axis through the origin, with constant, nonzero angular velocity. Let θ denote the angle swept out from the initial position, measured in the direction of rotation. The constant vector $\boldsymbol{\omega}$ indicates the axis and the sense of rotation; its magnitude is the angular speed, that is, the angle swept out per unit time:

$$\frac{d\theta}{dt} = \|\boldsymbol{\omega}\|. \quad (26)$$

Fix a total time t and divide it into n equal intervals of duration $\Delta t = t/n$; thus, $n\Delta t = t$. Since the angular speed is constant, the angle swept out in each interval is

$$\Delta\theta = \|\boldsymbol{\omega}\| \Delta t. \quad (27)$$

The n rotations occur about the same axis and in the same direction, so their angles add. Substituting the preceding expression and using $n\Delta t = t$, we obtain the total angle:

$$\theta = n\Delta\theta = n(\|\boldsymbol{\omega}\| \Delta t) = \|\boldsymbol{\omega}\| \underbrace{(n\Delta t)}_t = \|\boldsymbol{\omega}\| t. \quad (28)$$

More steps do not mean more time or a larger angle: as n increases, each step takes less time and rotates less, but t and θ remain fixed. The total angle can be large; see Figure 1.

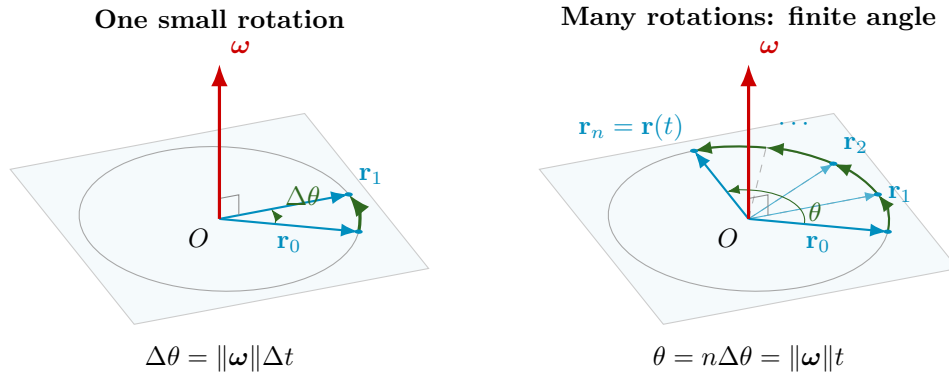


Figure 1. Small rotations accumulating into a finite rotation, shown in perspective in $\mathbb{R}^3$. The red vector $\boldsymbol{\omega}$ is perpendicular to the plane of rotation. The blue vectors $\mathbf{r}_k = \mathbf{r}(k\Delta t)$ show successive exact positions in that plane ($\mathbf{r}_0 \perp \boldsymbol{\omega}$), up to $\mathbf{r}_n = \mathbf{r}(t)$. As n increases, $\Delta\theta$ decreases, but $\theta = n\Delta\theta$ is preserved.

How does each small rotation translate into the motion of the point? In the case shown, $\mathbf{r} \perp \boldsymbol{\omega}$, so the radius of the circle is $\|\mathbf{r}\|$. When rotating through an angle $\Delta\theta$, measured in radians, the point travels along an arc of length $\|\mathbf{r}\| \Delta\theta$. If the angle is small, the displacement $\Delta\mathbf{r}$ joining the initial and final positions can be approximated by a vector tangent to the circle, of that same length. This is the first-order approximation.

The cross product $\boldsymbol{\omega} \times \mathbf{r}$ points precisely in that tangential direction and in the sense of rotation. Its magnitude is $\|\boldsymbol{\omega}\| \|\mathbf{r}\|$; therefore, multiplying it by Δt and using equation (27), we obtain the length we need:

$$\|\Delta t(\boldsymbol{\omega} \times \mathbf{r})\| = \|\mathbf{r}\| \|\boldsymbol{\omega}\| \Delta t = \|\mathbf{r}\| \Delta\theta. \quad (29)$$

Dividing the displacement by Δt and taking the limit, we identify $\boldsymbol{\omega} \times \mathbf{r}$ with the **instantaneous linear velocity of the point**:

$$\mathbf{v}(t) := \frac{d\mathbf{r}(t)}{dt} = \boldsymbol{\omega} \times \mathbf{r}(t) = \hat{\boldsymbol{\omega}}\mathbf{r}(t). \quad (30)$$

Thus, during a short interval, **the displacement is approximately the instantaneous linear velocity multiplied by the time**. The approximation is needed because the direction of the velocity changes as the point rotates. Write $\mathbf{r}_0 = \mathbf{r}(0)$ and $\mathbf{r}_1 := \mathbf{r}(\Delta t)$ for the positions before and after the first interval. The displacement is their difference; adding it to $\mathbf{r}_0$ gives the new position:

$$\begin{aligned} \Delta\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_0 &\approx \mathbf{v}(0) \Delta t = \Delta t \hat{\boldsymbol{\omega}}\mathbf{r}_0, \\ \mathbf{r}_1 &\approx \mathbf{r}_0 + \Delta t \hat{\boldsymbol{\omega}}\mathbf{r}_0 = (I + \Delta t \hat{\boldsymbol{\omega}})\mathbf{r}_0. \end{aligned} \quad (31)$$

Here I is the identity matrix, and the step only approximates the rotation. The formula also holds for any $\mathbf{r}$: its component parallel to the axis remains fixed and does not contribute to the cross product.

Each new position serves as the starting point for the next step. Write $\mathbf{r}_k := \mathbf{r}(k\Delta t)$ for the successive positions, as in Figure 1. Repeating the approximation gives

$$\begin{aligned} \mathbf{r}_1 &\approx (I + \Delta t \hat{\boldsymbol{\omega}})\mathbf{r}_0, \\ \mathbf{r}_2 &\approx (I + \Delta t \hat{\boldsymbol{\omega}})\mathbf{r}_1 \approx (I + \Delta t \hat{\boldsymbol{\omega}})^2\mathbf{r}_0, \\ &\vdots \\ \mathbf{r}_n &\approx (I + \Delta t \hat{\boldsymbol{\omega}})^n\mathbf{r}_0. \end{aligned} \quad (32)$$

The power n appears because we apply the same step n times. Since $\Delta t = t/n$, we denote the approximate final position by

$$\mathbf{r}_n^{(n)} := \left(I + \frac{t}{n} \hat{\boldsymbol{\omega}}\right)^n \mathbf{r}_0. \quad (33)$$

As n increases, each step becomes smaller and its approximation more accurate. The matrix limit defining the exponential gives

$$\lim_{n \rightarrow \infty} \left(I + \frac{t}{n} \hat{\boldsymbol{\omega}}\right)^n \mathbf{r}_0 = e^{t\hat{\boldsymbol{\omega}}} \mathbf{r}_0. \quad (34)$$

Thus, the accumulation of infinitesimal rotations produces a finite rotation, described by the exponential. While each angle $\Delta\theta$ tends to zero, the total angle $\theta = \|\boldsymbol{\omega}\|t$ remains fixed. The exact position is

$$\mathbf{r}(t) := e^{t\hat{\boldsymbol{\omega}}} \mathbf{r}_0. \quad (35)$$

Differentiating this expression recovers the linear velocity in equation (30). To first order,

$$\mathbf{r}(t) = \mathbf{r}_0 + t(\boldsymbol{\omega} \times \mathbf{r}_0) + O(t^2), \quad (36)$$

which explains the name “infinitesimal generator.”

3.3 The bracket is not associative

The generators satisfy

$$[J_x, J_y] = J_z, \quad [J_y, J_z] = J_x, \quad [J_z, J_x] = J_y. \quad (37)$$

The triple J_x, J_y, J_z does not detect the failure of associativity, because

$$[[J_x, J_y], J_z] = [J_z, J_z] = 0 \quad (38)$$

and

$$[J_x, [J_y, J_z]] = [J_x, J_x] = 0. \quad (39)$$

The equality for this triple is accidental. Associativity would require equality for all triples. Taking J_x, J_x, J_y gives

$$[[J_x, J_x], J_y] = 0, \quad [J_x, [J_x, J_y]] = [J_x, J_z] = -J_y. \quad (40)$$

Therefore,

$$\boxed{[[J_x, J_x], J_y] \neq [J_x, [J_x, J_y]]}. \quad (41)$$

Warning

Matrix multiplication AB is indeed associative. The bracket $[A, B] = AB - BA$, constructed from it, generally is not. There is no contradiction: these are two different operations on related objects.

4 The tensor algebra: associative, free, and noncommutative

Let V be a vector space. Its tensor algebra is

$$T(V) := \bigoplus_{n=0}^{\infty} V^{\otimes n} = \mathbb{K} \oplus V \oplus (V \otimes V) \oplus (V^{\otimes 3}) \oplus \dots \quad (42)$$

Multiplication is concatenation:

$$\begin{aligned} (u_1 \otimes \dots \otimes u_m)(v_1 \otimes \dots \otimes v_n) \\ := u_1 \otimes \dots \otimes u_m \otimes v_1 \otimes \dots \otimes v_n. \end{aligned} \quad (43)$$

This multiplication is already associative. If we group three tensor words, the final concatenation is the same:

$$(ab)c = a(bc) = abc. \quad (44)$$

If V has basis $x_1, \dots, x_n$, then

$$T(V) \cong \mathbb{K}\langle x_1, \dots, x_n \rangle, \quad (45)$$

the free associative algebra, or algebra of noncommutative polynomials. In it, $x_i x_j$ and $x_j x_i$ are different words before relations are imposed.

4.1 Three similar notations, three different objects

Before continuing, let us distinguish this notation from two others that should not be confused. The expressions

$$\mathbb{K}\langle x_1, \dots, x_n \rangle, \quad \mathbb{K}[x_1, \dots, x_n], \quad \mathbb{K}(x_1, \dots, x_n) \quad (46)$$

look similar but represent different objects.

4.1.1 Angle brackets: noncommutative polynomials

$$\mathbb{K}\langle x_1, \dots, x_n \rangle \quad (47)$$

is the free associative algebra. Its monomials are words and, in general,

$$x_i x_j \neq x_j x_i. \quad (48)$$

For two generators, part of its basis is

$$1, x, y, x^2, xy, yx, y^2, x^3, x^2y, xyx, yx^2, \dots \quad (49)$$

4.1.2 Square brackets: commutative polynomials

$$\mathbb{K}[x_1, \dots, x_n] \quad (50)$$

is the usual polynomial algebra, where

$$x_i x_j = x_j x_i. \quad (51)$$

It can be obtained as a quotient of the free algebra:

$$\mathbb{K}[x_1, \dots, x_n] \cong \frac{\mathbb{K}\langle x_1, \dots, x_n \rangle}{\langle x_i x_j - x_j x_i \rangle}. \quad (52)$$

4.1.3 Parentheses: rational functions

$$\mathbb{K}(x_1, \dots, x_n) \quad (53)$$

is the field of fractions of $\mathbb{K}[x_1, \dots, x_n]$. Its elements are quotients

$$\frac{p(x_1, \dots, x_n)}{q(x_1, \dots, x_n)}, \quad q \neq 0. \quad (54)$$

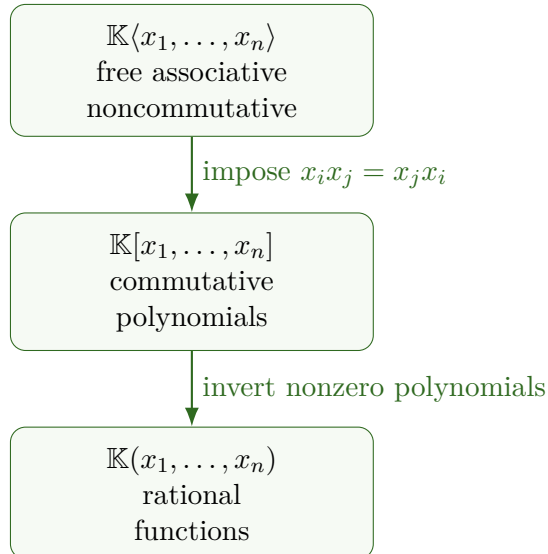


Figure 2. From the free associative algebra to commutative polynomials and their field of fractions.

4.2 Universal property of the tensor algebra

Let us now return to $T(V)$. To express precisely in what sense it is free, recall which maps preserve the structure of a unital associative algebra.

Definition 3 (Homomorphism of unital associative algebras). Let A and B be unital associative algebras over the same field $\mathbb{K}$. A map $F : A \rightarrow B$ is a *homomorphism of unital associative algebras* if it is $\mathbb{K}$ -linear, preserves the product, and preserves the unit. Explicitly, for all $\lambda, \mu \in \mathbb{K}$ and $a, b \in A$, it must satisfy

$$\begin{aligned} F(\lambda a + \mu b) &= \lambda F(a) + \mu F(b), & (\text{linearity}), \\ F(ab) &= F(a)F(b), & (\text{preservation of the product}), \\ F(1_A) &= 1_B, & (\text{preservation of the unit}). \end{aligned} \tag{55}$$

Property 1 (Universal property of the tensor algebra $T(V)$). Let $j_V : V \rightarrow T(V)$ be the canonical inclusion of V as the component $V^{\otimes 1}$. For every linear map $f : V \rightarrow A$ into a unital associative algebra A , there exists a unique homomorphism of unital associative algebras $\tilde{f} : T(V) \rightarrow A$ such that $f = \tilde{f} \circ j_V$.

$$\begin{array}{ccc} V & \xrightarrow{j_V} & T(V) \\ & \searrow f & \swarrow \tilde{f} \\ & A & \end{array}$$

$f = \tilde{f} \circ j_V, \quad \tilde{f} \text{ is unique.}$

Figure 3. Commutative diagram of the universal property of the tensor algebra $T(V)$.

The word “free” means precisely that no additional relations are imposed among the generators, apart from linearity, the unit, and associativity.

Warning

$T(V)$ is the free unital associative algebra generated by V . It is not the free nonassociative algebra: in the latter, $(xy)z$ and $x(yz)$ would be distinct elements, whereas in $T(V)$ they are equal. Thus, the freeness of $T(V)$ is understood within the category of unital associative algebras.

5 Construction of the universal enveloping algebra

Now let $\mathfrak{g}$ be a Lie algebra. We start with the free associative algebra $T(\mathfrak{g})$ and require the associative commutator to reproduce the Lie bracket. For all $x, y \in \mathfrak{g}$, we impose

$$xy - yx = [x, y]_{\mathfrak{g}}. \tag{56}$$

Definition 4 (Universal enveloping algebra). The universal enveloping algebra of $\mathfrak{g}$ is

$$U(\mathfrak{g}) := \frac{T(\mathfrak{g})}{\langle x \otimes y - y \otimes x - [x, y]_{\mathfrak{g}} \mid x, y \in \mathfrak{g} \rangle}. \tag{57}$$

The denominator is the two-sided ideal generated by these relations.

The quotient does not impose associativity: it was already present in $T(\mathfrak{g})$. What it imposes is compatibility between associative multiplication and the original bracket.

Key idea

Crucial point. The product $x \otimes y$ in $T(\mathfrak{g})$ is not identified with the bracket $[x, y]_{\mathfrak{g}}$. When forming $U(\mathfrak{g}) = T(\mathfrak{g})/I$, the ideal I imposes $x \otimes y - y \otimes x - [x, y]_{\mathfrak{g}} = 0$. Let $q : T(\mathfrak{g}) \twoheadrightarrow U(\mathfrak{g})$ be the canonical projection and let $\iota = q|_{\mathfrak{g}}$. Since $q(a) = a + I$, in the quotient we directly have

$$\begin{aligned} [\iota(x), \iota(y)]_{U(\mathfrak{g})} &:= \iota(x)\iota(y) - \iota(y)\iota(x) \\ &= (x + I)(y + I) - (y + I)(x + I) \\ &= x \otimes y - y \otimes x + I \\ &= [x, y]_{\mathfrak{g}} + I = \iota([x, y]_{\mathfrak{g}}). \end{aligned}$$

Thus, the image of the bracket of $\mathfrak{g}$ coincides in $U(\mathfrak{g})$ with the class of the commutator of the generators x and y inside the tensor algebra. **It is the commutator, not the product, that reproduces the bracket**, and this equality appears only after passing to the quotient.

Example 1 (Presentation using a basis). If $x_1, \dots, x_n$ is a basis of $\mathfrak{g}$, then the free noncommutative associative algebra generated by these elements is $T(\mathfrak{g}) \cong \mathbb{K}\langle x_1, \dots, x_n \rangle$. Therefore,

$$U(\mathfrak{g}) \cong \frac{\mathbb{K}\langle x_1, \dots, x_n \rangle}{\langle x_i x_j - x_j x_i - [x_i, x_j]_{\mathfrak{g}} \rangle}. \quad (58)$$

5.1 Associativity and commutativity

Proposition 2 (Associativity of a quotient). *Let A be a unital associative algebra and let $I \subseteq A$ be a two-sided ideal. Then the quotient A/I , with multiplication*

$$(a + I)(b + I) := ab + I. \quad (59)$$

is an associative algebra with unit $1 + I$.

Proof of Proposition 2. First we must check that the product does not depend on the representatives. If $a' = a + i$ and $b' = b + j$, with $i, j \in I$, then

$$a'b' - ab = (a + i)(b + j) - ab = aj + ib + ij \in I, \quad (60)$$

because I is two-sided. Thus, $a'b' + I = ab + I$, and multiplication of classes is well defined. Now, using the associativity of A ,

$$\begin{aligned} ((a + I)(b + I))(c + I) &= (ab + I)(c + I) = (ab)c + I \\ &= a(bc) + I = (a + I)(bc + I) \\ &= (a + I)((b + I)(c + I)). \end{aligned} \quad (61)$$

Moreover, $1 + I$ is the unit of the quotient. □

Applying Proposition 2 to $A = T(\mathfrak{g})$ and the two-sided ideal in equation (57), we conclude that $U(\mathfrak{g}) = T(\mathfrak{g})/I$ is a unital associative algebra.

However, $U(\mathfrak{g})$ is not commutative in general. The relation defining the quotient says exactly that

$$xy - yx = [x, y]_{\mathfrak{g}}. \quad (62)$$

Thus, the possible failure of commutativity in $U(\mathfrak{g})$ preserves the information in the original bracket. For example, in $U(\mathfrak{so}(3))$ we will have

$$J_x J_y - J_y J_x = J_z \neq 0. \quad (63)$$

If $\mathfrak{g}$ is abelian, then $[x, y]_{\mathfrak{g}} = 0$ for all x, y , and the relations impose $xy = yx$. In this case, the enveloping algebra is the symmetric algebra of $\mathfrak{g}$:

$$S(\mathfrak{g}) := \frac{T(\mathfrak{g})}{\langle x \otimes y - y \otimes x : x, y \in \mathfrak{g} \rangle}, \quad U(\mathfrak{g}) \cong S(\mathfrak{g}). \quad (64)$$

If $\mathfrak{g}$ has a finite basis $x_1, \dots, x_n$, this identification is written as

$$U(\mathfrak{g}) \cong \mathbb{K}[x_1, \dots, x_n]. \quad (65)$$

This is precisely the quotient by the commutativity relations described when distinguishing the three notations in equation (52). Conversely, the Poincaré-Birkhoff-Witt theorem^[2] allows us to identify $\mathfrak{g}$ with its image inside $U(\mathfrak{g})$. If $U(\mathfrak{g})$ were commutative, then $xy - yx = 0$ would imply $[x, y]_{\mathfrak{g}} = 0$. Therefore,

$$\boxed{U(\mathfrak{g}) \text{ is commutative} \iff \mathfrak{g} \text{ is abelian.}} \quad (66)$$

Key idea

$U(\mathfrak{g})$ is always associative and unital, but it is commutative only when $\mathfrak{g}$ is abelian. For a nonabelian Lie algebra, the noncommutativity of $U(\mathfrak{g})$ is not a defect: it is the mechanism that remembers the bracket.

5.2 Universal property

Proposition 3 (The commutator defines a Lie algebra). *Let A be an associative algebra over $\mathbb{K}$, with or without a unit. On the underlying vector space of A , define*

$$[a, b]_{A_{\text{Lie}}} := ab - ba. \quad (67)$$

Then this bracket is bilinear, alternating, and satisfies the Jacobi identity. It therefore defines a Lie algebra, denoted by A_{Lie} .

Proof of Proposition 3. Bilinearity follows from the bilinearity of the product of A . Moreover,

$$[a, a]_{A_{\text{Lie}}} = aa - aa = 0, \quad (68)$$

so the bracket is alternating. For Jacobi, associativity allows us to omit parentheses and expand

$$\begin{aligned} & [a, [b, c]] + [b, [c, a]] + [c, [a, b]] \\ &= (abc - acb - bca + cba) \\ &+ (bca - bac - cab + acb) \\ &+ (cab - cba - abc + bac) \\ &= 0. \end{aligned} \quad (69)$$

^[2]**Poincaré–Birkhoff–Witt theorem.** Let $\mathfrak{g}$ be a Lie algebra over $\mathbb{K}$ and let $(x_i)_{i \in I}$ be a totally ordered basis of $\mathfrak{g}$. Then $\iota : \mathfrak{g} \hookrightarrow U(\mathfrak{g})$ is injective—hence, $\mathfrak{g} \cong \iota(\mathfrak{g}) \subset U(\mathfrak{g})$ —and the ordered monomials

$$\iota(x_{i_1}) \cdots \iota(x_{i_r}), \quad r \geq 0, \quad i_1 \leq \cdots \leq i_r, \quad (I)$$

form a $\mathbb{K}$ -basis of $U(\mathfrak{g})$; for $r = 0$, the monomial is understood to be 1. It will not be proved here.

Each monomial appears once with a positive sign and once with a negative sign; therefore, the Jacobi identity holds. $\square$

Definition 5 (Homomorphism of Lie algebras). Let $\mathfrak{g}$ and $\mathfrak{h}$ be Lie algebras over the same field $\mathbb{K}$. A map $f : \mathfrak{g} \rightarrow \mathfrak{h}$ is a *homomorphism of Lie algebras* if it is $\mathbb{K}$ -linear and preserves the Lie product, that is, the bracket. Explicitly, for all $\lambda, \mu \in \mathbb{K}$ and $x, y \in \mathfrak{g}$, it must satisfy

$$\begin{aligned} f(\lambda x + \mu y) &= \lambda f(x) + \mu f(y), & (\text{linearity}), \\ f([x, y]_{\mathfrak{g}}) &= [f(x), f(y)]_{\mathfrak{h}}, & (\text{preservation of the Lie product}). \end{aligned} \quad (70)$$

The following proposition shows that every homomorphism of Lie algebras $f : \mathfrak{g} \rightarrow A_{\text{Lie}}$ admits a unique lift to a homomorphism of unital associative algebras $\tilde{f} : U(\mathfrak{g}) \rightarrow A$. Here “lift” means that the corresponding triangle commutes, that is, $\tilde{f} \circ \iota = f$.

Proposition 4 (Universal property of the enveloping algebra). *Let A be a unital associative algebra, let $\iota : \mathfrak{g} \rightarrow U(\mathfrak{g})$ be the canonical map, and let $f : \mathfrak{g} \rightarrow A_{\text{Lie}}$ be a homomorphism of Lie algebras. Then there exists a unique homomorphism of unital associative algebras*

$$\tilde{f} : U(\mathfrak{g}) \longrightarrow A \quad (71)$$

such that $\tilde{f} \circ \iota = f$. Equivalently, assigning to f its unique extension defines a natural bijection

$$\Phi_A : \text{Hom}_{\text{LieAlg}}(\mathfrak{g}, A_{\text{Lie}}) \xrightarrow{\sim} \text{Hom}_{\text{AssocAlg}}(U(\mathfrak{g}), A), \quad f \longmapsto \tilde{f}. \quad (72)$$

Its inverse is

$$\Psi_A : \text{Hom}_{\text{AssocAlg}}(U(\mathfrak{g}), A) \longrightarrow \text{Hom}_{\text{LieAlg}}(\mathfrak{g}, A_{\text{Lie}}), \quad \tilde{f} \longmapsto \tilde{f} \circ \iota. \quad (73)$$

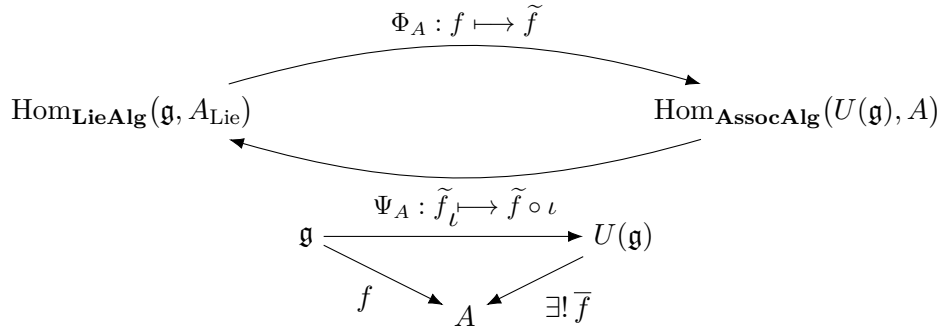


Figure 4. The natural bijection Φ_A in equation (72) and its inverse Ψ_A . The lower triangle expresses that there exists a unique extension $\tilde{f}$ of f , where $\tilde{f} = \tilde{f}$, and that $\tilde{f} \circ \iota = f$.

Proof of Proposition 4. Since f is linear, the universal property of $T(\mathfrak{g})$ provides a unique homomorphism of unital associative algebras $\hat{f} : T(\mathfrak{g}) \rightarrow A$ extending f . For all $x, y \in \mathfrak{g}$,

$$\begin{aligned} \hat{f}(x \otimes y - y \otimes x - [x, y]_{\mathfrak{g}}) &= f(x)f(y) - f(y)f(x) - f([x, y]_{\mathfrak{g}}) \\ &= [f(x), f(y)]_{A_{\text{Lie}}} - f([x, y]_{\mathfrak{g}}) \\ &= f([x, y]_{\mathfrak{g}}) - f([x, y]_{\mathfrak{g}}) = 0, \end{aligned} \quad (74)$$

because f preserves the Lie bracket and the bracket of A_{Lie} is the commutator. Since $\ker \hat{f}$ is a two-sided ideal, the preceding equation implies

$$x \otimes y - y \otimes x - [x, y]_{\mathfrak{g}} \in \ker \hat{f} \quad \text{for all } x, y \in \mathfrak{g}, \quad \implies \quad I \subseteq \ker \hat{f}. \quad (75)$$

The homomorphism $\hat{f}$ then factors through the quotient and yields a homomorphism $\tilde{f} : U(\mathfrak{g}) = T(\mathfrak{g})/I \rightarrow A$ satisfying $\tilde{f} \circ \iota = f$.

For uniqueness, let $q : T(\mathfrak{g}) \rightarrow U(\mathfrak{g})$ be the canonical projection. If $h : U(\mathfrak{g}) \rightarrow A$ also extends f , then $h \circ q : T(\mathfrak{g}) \rightarrow A$ is a homomorphism of unital associative algebras extending f . By uniqueness of $\hat{f}$, we have $h \circ q = \hat{f} = \tilde{f} \circ q$. Since q is surjective, $h = \tilde{f}$.

The existence and uniqueness just proved therefore define the map

$$\Phi_A : \text{Hom}_{\text{LieAlg}}(\mathfrak{g}, A_{\text{Lie}}) \longrightarrow \text{Hom}_{\text{AssocAlg}}(U(\mathfrak{g}), A), \quad \Phi_A(f) := \tilde{f}. \quad (76)$$

In the opposite direction, if $F : U(\mathfrak{g}) \rightarrow A$ is a homomorphism of unital associative algebras, then $F \circ \iota : \mathfrak{g} \rightarrow A_{\text{Lie}}$ is a homomorphism of Lie algebras. This defines

$$\Psi_A(F) := F \circ \iota. \quad (77)$$

Now,

$$\Psi_A(\Phi_A(f)) = \tilde{f} \circ \iota = f. \quad (78)$$

Conversely, F is an extension of $F \circ \iota$; by the uniqueness already proved, that extension is precisely F . Consequently,

$$\Phi_A(\Psi_A(F)) = F. \quad (79)$$

Thus, Φ_A and Ψ_A are inverses, establishing the natural bijection in equation (72).

□

Thus, the universal enveloping functor is

$$U : \text{LieAlg}_{\mathbb{K}} \longrightarrow \text{AssocAlg}_{\mathbb{K}}, \quad (80)$$

and it is left adjoint to the commutator functor $A \mapsto A_{\text{Lie}}$.

Key idea

The universal enveloping functor does not go from groups to Lie algebras. The passage from a Lie group to its algebra is $G \mapsto T_e G$. The passage from a Lie algebra to an associative algebra is $\mathfrak{g} \mapsto U(\mathfrak{g})$.

6 Commutators, derivations, and quantum mechanics

In an abstract Lie algebra, only the bracket is defined. In particular, if $X, Y \in \mathfrak{g}$, the expression XY has no intrinsic meaning, and therefore neither does $[XY, Z]$. This expression becomes meaningful when we work in an ambient associative algebra, for example in $U(\mathfrak{g})$ or in an algebra of operators.

Proposition 5 (Leibniz rule for the commutator). *Let A be a unital associative algebra and consider the commutator bracket $[X, Y] := XY - YX$. For all $X, Y, Z \in A$, we have*

$$[XY, Z] = X[Y, Z] + [X, Z]Y, \quad (81)$$

$$[X, YZ] = [X, Y]Z + Y[X, Z]. \quad (82)$$

Proof of Proposition 5. Adding and subtracting XZY and using associativity,

$$\begin{aligned}
[XY, Z] &= XYZ - ZXY \\
&= XYZ - XZY + XZY - ZXY \\
&= X(YZ - ZY) + (XZ - ZX)Y \\
&= X[Y, Z] + [X, Z]Y.
\end{aligned} \tag{83}$$

The second identity is obtained in the same way, by adding and subtracting YXZ .

□

This rule is not a consequence of the Jacobi identity: it is a direct consequence of associative multiplication. The Jacobi identity also follows from the associative commutator and can be written as

$$[X, [Y, Z]] = [[X, Y], Z] + [Y, [X, Z]]. \tag{84}$$

The conceptual difference is important: equations (81) and (82) describe how the commutator acts on the *associative product*, whereas equation (84) describes how it acts on another bracket.

Conceptual bridge

In quantum mechanics, one works with a noncommutative associative algebra of operators on a Hilbert space. Observables correspond to self-adjoint operators, and their Lie product is usually normalized as

$$[A, B]_{\text{q}} := \frac{1}{i\hbar}[A, B]. \tag{85}$$

The Heisenberg equation for an observable A is

$$\frac{dA}{dt} = \frac{i}{\hbar}[H, A] + \frac{\partial A}{\partial t}, \tag{86}$$

where H is the Hamiltonian. The Leibniz rule ensures that the evolution generated by the commutator with H respects products:

$$[H, AB] = [H, A]B + A[H, B]. \tag{87}$$

Thus, the passage $\mathfrak{g} \rightarrow U(\mathfrak{g})$ is not merely formal: it provides the associative product needed to compose operators and, at the same time, recovers the Lie bracket as a commutator. The product of two observables need not be self-adjoint again, although it does exist in the operator algebra. In applications involving unbounded operators, their common domains must also be carefully handled; here we reason at the algebraic level.

7 From the Lie group to the universal enveloping algebra

Now that $U(\mathfrak{g})$ has been defined, we can assemble into a single chain the passage from a Lie group to an associative algebra. A Lie group G has two compatible structures:

- it is a differentiable manifold;
- it is a group whose multiplication and inversion are smooth maps.

The group multiplication

$$G \times G \longrightarrow G, \quad (g, h) \longmapsto gh, \quad (88)$$

is indeed associative:

$$(gh)k = g(hk). \quad (89)$$

The associated Lie algebra is the tangent space at the identity,

$$\text{Lie}(G) := T_e G. \quad (90)$$

Each vector in $T_e G$ determines a left-invariant vector field. The bracket of $\text{Lie}(G)$ is obtained from the bracket of these vector fields.

Conceptual bridge

Associativity resides in the finite product gh of the group. When passing to infinitesimal generators, the natural operation is no longer that product, but the bracket. It satisfies Jacobi and, in general, is not associative.

Applying the enveloping construction to $\mathfrak{g} = \text{Lie}(G)$ gives the chain:

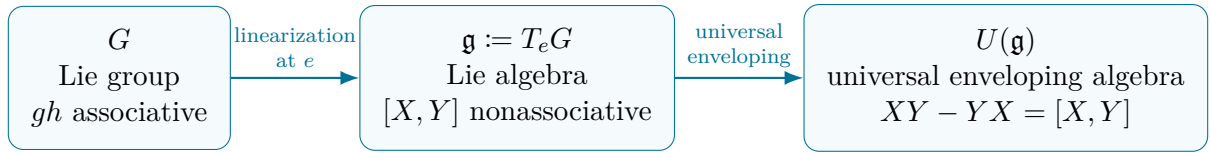


Figure 5. From the Lie group to its Lie algebra and then to the universal enveloping algebra.

Main properties of $U(\mathfrak{g})$. We collect the properties established in the preceding sections:

- $U(\mathfrak{g})$ is a unital associative $\mathbb{K}$ -algebra; in general, it is not commutative.
- There is a canonical map $\iota : \mathfrak{g} \rightarrow U(\mathfrak{g})$ satisfying

$$\iota([X, Y]_{\mathfrak{g}}) = \iota(X)\iota(Y) - \iota(Y)\iota(X). \quad (91)$$

- Universal property: every Lie homomorphism from $\mathfrak{g}$ to the Lie algebra underlying an associative algebra A extends uniquely to a homomorphism of unital associative algebras $U(\mathfrak{g}) \rightarrow A$; see Figure 4.
- The Poincaré-Birkhoff-Witt theorem implies that ι is injective and that the ordered monomials in a basis of $\mathfrak{g}$ form a basis of $U(\mathfrak{g})$.
- $U(\mathfrak{g})$ is commutative if and only if $\mathfrak{g}$ is abelian.

8 Example: the enveloping algebra of the two-dimensional algebra

Let us return to

$$\mathfrak{g} = \text{span}\{x, y\}, \quad [x, y] = y. \quad (92)$$

Its enveloping algebra is

$$U(\mathfrak{g}) = \frac{\mathbb{K}\langle x, y \rangle}{\langle xy - yx - y \rangle}. \quad (93)$$

Inside $U(\mathfrak{g})$,

$$xy = yx + y. \quad (94)$$

This allows us to reorder words. For example,

$$\begin{aligned} xy^2 &= (xy)y \\ &= (yx + y)y \\ &= yxy + y^2 \\ &= y(yx + y) + y^2 \\ &= y^2x + 2y^2. \end{aligned} \quad (95)$$

By induction,

$$xy^n = y^n x + ny^n. \quad (96)$$

The Poincaré-Birkhoff-Witt theorem guarantees that, after fixing an order, the ordered monomials form a basis. With the order $y < x$, a basis is

$$\{y^a x^b : a, b \geq 0\}. \quad (97)$$

9 The enveloping algebra of $\mathfrak{so}(3)$

Using the generators J_x, J_y, J_z , we obtain

$$U(\mathfrak{so}(3)) = \frac{\mathbb{R}\langle J_x, J_y, J_z \rangle}{I}, \quad (98)$$

where I is the two-sided ideal generated by

$$J_x J_y - J_y J_x - J_z, \quad (99)$$

$$J_y J_z - J_z J_y - J_x, \quad (100)$$

$$J_z J_x - J_x J_z - J_y. \quad (101)$$

For example,

$$J_x J_y = J_y J_x + J_z. \quad (102)$$

Associative expressions of arbitrary degree now make sense, such as

$$J_x^2, \quad J_z J_x J_y, \quad J_x^4 J_y^2. \quad (103)$$

The Poincaré-Birkhoff-Witt theorem states that, once the ordered basis $J_x < J_y < J_z$ of $\mathfrak{so}(3)$ is chosen, the ordered monomials form a basis of $U(\mathfrak{so}(3))$, and every element of the enveloping algebra is expressed uniquely as a finite sum of them:

$$\begin{aligned} \mathcal{B} &:= \{J_x^a J_y^b J_z^c : a, b, c \in \mathbb{Z}_{\geq 0}\} \text{ is a basis of } U(\mathfrak{so}(3)), \\ u &= \sum_{a,b,c \geq 0} \lambda_{abc} J_x^a J_y^b J_z^c, \quad \lambda_{abc} \in \mathbb{R}, \quad \lambda_{abc} = 0 \text{ except for finitely many indices.} \end{aligned} \quad (104)$$

The basis $\mathcal{B}$ is a basis of $U(\mathfrak{so}(3))$, not of $\mathfrak{so}(3)$. Products and powers are computed in $U(\mathfrak{so}(3))$, and $J_x^0 J_y^0 J_z^0 = 1$.

10 Final conceptual map

The rotation example provides a first connection: for constant angular velocity, each step approximates the next position, and composing many steps leads to the matrix exponential (34). This procedure reconstructs the finite motion; it is not the construction of the universal enveloping algebra.

The constructions discussed do not form a single operation, but a chain of steps of different kinds:

$$G \mapsto \text{Lie}(G) = T_e G \mapsto T(\text{Lie}(G)) \mapsto U(\text{Lie}(G)). \quad (105)$$

1. G is a Lie group. Its product gh is associative.
2. $\text{Lie}(G)$ is a vector space with a Lie bracket. The bracket is generally not associative, but it satisfies Jacobi.
3. $T(\text{Lie}(G))$ is the free associative algebra of noncommutative words on the infinitesimal generators.
4. $U(\text{Lie}(G))$ is the quotient imposing

$$XY - YX = [X, Y]. \quad (106)$$

Key idea

The essential subtlety is not to confuse three operations:

$$\boxed{\text{group product} \neq \text{Lie bracket} \neq \text{product in } U(\mathfrak{g})}. \quad (107)$$

The group product and the enveloping algebra product are associative. The bracket, in general, is not. The universal enveloping algebra relates them by realizing the bracket as a commutator.

11 Conclusion

Thinking of Lie algebras through infinitesimal transformations is correct and very useful, but it can obscure the difference between the associative group product and the infinitesimal bracket. The example of $SO(3)$ shows both at once: finite rotations compose associatively, while their generators J_x, J_y, J_z form a Lie algebra whose bracket is not associative. Repeating small steps explains how the exponential recovers a finite rotation.

The universal enveloping algebra solves the problem of passing from this infinitesimal language to an associative setting. It starts with the tensor algebra, which is already associative and free, and then imposes the relations $xy - yx = [x, y]$. Thus, **the original bracket is recovered as a commutator, not as a product**, within a unital associative algebra, and the notation of noncommutative polynomials appears naturally. This distinction also explains the Leibniz rule for the commutator used in quantum mechanics.